

Alahe Akhavan

 akhavana |  alaheakhavan |  Portfolio |  alahe.akhavan@gmail.com |  (279) 758-1676

EDUCATION

Johns Hopkins University

Master of Science in Mechanical Engineering

Berkeley, CA

August 2026 – May 2028

University of California, Berkeley

Bachelor of Science in Mechanical Engineering

Berkeley, CA

July 2023 – December 2025

EXPERIENCE

Assistive Grasping, Research Assistant

Embodied Dexterity Group (EDG), Advised by Prof. Hannah Stuart

May 2024 – June 2025

Berkeley, CA

- Designed and prototyped a next-generation Dorsal Grasper for C6/C7 SCI, improving grasp stability and enabling functional object manipulation in human trials.
- Designed integrated electromechanical system including custom PCB, and 3D-printed structure for rapid prototyping and iterative testing.
- Contributed in human-subject testing across participants with varying SCI levels and, manuscript currently under peer review.

Tremor Suppression Glove (Biomedical Device), Undergrad Research

Advised by Prof. Grace O'Connell

August 2024 – July 2025

Berkeley, CA

- Designed a wearable assistive device to attenuate pathological tremors using mechanical stabilization and actuation strategies.
- Prototyped and tested multiple design iterations, balancing effectiveness, wearability, and user comfort.
- Collaborated with a team of engineers, gained co-design, biomedical device participant interviewing and testing.

Prosthetic Hand with Dexterous Articulation for Grasping and Manipulation, Research Fellow

Assistive Robotics and Manipulation Lab (ARMLab), Advised by Prof. Monroe Kennedy III

June 2025 – August 2025

Stanford, CA

- Developed a control framework for dexterous robotic manipulation using multimodal sensing (vision + biosignal input).
- Implemented real-time control strategies for articulated object manipulation, improving robustness in articulated tasks.
- Integrated sensing, perception, and control pipelines within a robotic system for closed-loop interaction.

Tactile Sensing Control Optimization, Research Assistant

Embodied Dexterity Group (EDG), Advised by Prof. Hannah Stuart

January 2024 – April 2024

Berkeley, CA

- Optimized the Universal Robot 10 control system for path planning and to reduce oscillations.
- Implemented and tested control strategies using ROS, C++, and python languages.
- Enhanced robotic control workflows with RViz and Gazebo simulations.

Multi-robot Collaborative Planning for Package Transportation, Research Fellow June 2022 – August 2022

Berkeley Artificial Intelligence Research (BAIR) Lab, Advised by Prof. Claire J. Tomline

Berkeley, CA

- Designed a motion planning control system for a multi-robot research project, using Universal Robot 5 (UR5) and Quadruped Robot to transport packages in a warehouse setting.
- Created over 6 motion stages for multi-robot package delivery tasks with the UR5.
- Worked with Python in ROS, and tested tasks using simulations RViz and Gazebo to see real time motion and map environments.

Extended Opportunity Programs and Services, STEM Tutor

January 2022 – May 2023

American River College

Sacramento, CA

- Helped students in Math, Physics, and Computer Sciences across diverse and different learning styles.
- Gained leadership and communication skills through working with diverse community of students from all backgrounds.

PROJECTS

Flexible Wearable Garment

January 2025 – May 2025

Product Development Research Project

Berkeley, CA

- Worked with a team of Ph.D. engineering students to develop a transformable textile that uses smock patterns to expand over 200% for programmable local and global adaptability to body geometry with tunable structural stability and breathability.
- We presented the design of the dress for pregnant women to potential users who have experienced pregnancy, aiming to collect their feedback on our design.
- Our product shows promise in applications across sportswear, children's clothing, and adjustable fashion accessories, thanks to its versatility and mechanical programmability.

Tea Making Machine, Mechanical Engineering Capstone Project

August 2024 – December 2024

Senior Capstone Project

Berkeley, CA

- Worked with a team of engineering students to create a fully functional traditional tea making machine for our capstone project in Mechatronics course.
- Contributed in creating the CAD design for tea dispensing, testing electronics, PCB design, and programming via C++.
- Our team won first place for the electromechanical award.

Maze Navigator Robot

January 2024 – May 2024

Personal Project

Berkeley, CA

- Designed and implemented an IoT-based path navigation system using a rover robot to autonomously follow maze-like environments while monitoring surroundings.
- Optimized ultrasonic sensor response (Adafruit) for accurate boundary detection.
- Developed a wireless communication system (Wi-Fi) to transmit real-time sensor data, distance measurements, and path duration during navigation.

Biomedical Assistive Device for Parkinson's

September 2023 – February 2024

mentored by Prof. Hannah Stuart

Berkeley, CA

- Designing a fully co-designed mechanical spoon to reduce hand tremor for people with Parkinson's.
- Material testing and 3D manufacturing; mechanical testing machines; 3D modeling in SolidWorks and Fusion.

A* Pathfinder Navigation in Different Environments

August 2023 – December 2023

Mentored by Dr. Jason J. Choi

Sacramento, CA

- Developed an A* (Astar) path finding algorithm for a Unitree Go1 robot.
- Designed multiple simulated environments to test the algorithm, tested blocks and different path types.
- Improved skills in robotic programming, algorithm design, testing and simulation skills; learned how to utilize ROS and Gazebo.

PUBLICATIONS

Evaluating Supernumerary Dorsal Grasping in the Home for People with C5–C7 Spinal Cord Injury

Andrew I.W. McPherson, Jungpyo Lee, **Alahe Akhavan**, Hannah S. Stuart.

IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) (Submitted for review)

Assistive Dorsal Grasper Modifications for In-Home Experiments

Alahe Akhavan, Andrew I.W. McPherson, Jungpyo Lee, Hannah S. Stuart.

Poster Presentation, SACNAS National Diversity in STEM Conference, 2024.

SERVICE/LEADERSHIP

Undergraduate Mentor

January 2024 – present

- Mentored incoming and transfer students to support their transition into UC Berkeley.
- Introduced mentees to university resources and encouraged participation in networking opportunities.
- Provided guidance on academic decisions while fostering confidence, motivation, and personal growth.

STEM Mentor and Volunteer

January 2025 – March 2025

- Mentored a group of diverse middle school students for their first scientific research experiment.
- Advocate for STEM through motivating research projects that are designed and completed by students.
- Helped young students gain communication, research, data collection, and analysis skills.

STEM Outreach

August 2024 – May 2025

- As a member of EDG Lab, partnered with disability organizations and managed information booths in the resource fairs to showcase past projects
- Organized a skill workshop on assistive devices for engineering students to gain manufacturing design, sensor testing, and circuit board design.

STEM Tutor

January 2022 – May 2023

- At my local community college, assisted people across a range of age groups and learning styles, helping them with STEM courses.
- Helped create workshops and regular check-ins to ensure and encourage student progress.

Volunteer Tutor

January 2017 – December 2021

- Thought and tutored low-income students of all ages with math and language at a Sunday school.

MENTORSHIP

Current Students

Jeremiah Avitia, Cristo Alberto Barragan, Gustavo Inga, Enkhjin Erdenee, September 2025 – present
Tessa Carol Brieger

Past Students

Cathy Sheng, Cassie Katherine Ly, Joya Mirabai Stewart, Alicia Wang January 2024 – Dec 2024

CONFERENCES

ASME IMECE2024 Conference Portland, Oregon November 2024

- Poster Presentation: Assistive Dorsal Grasper Modifications for In-Home experiments

SACNAS Conference Arizona State University, Phoenix, Arizona October 2024

- Poster Presentation: Assistive Dorsal Grasper Modifications for In-Home experiments

Undergraduate Women in Physics APS University of California, Santa Cruz December 2022

Silicon Valley Women in Engineering Conference Virtual March 2022

AWARDS

AUTODESK Instructables Award for Best Student Project (TremorEz) September 2025

The Leadership Award TLA May 2025

SACNAS National Diversity in STEM Conference Travel Scholarship June 2024

UC Berkeley Mechanical Engineering Departmental Award August 2023